

NASSAU CANDY DISTRIBUTOR

Profit Intelligence & Margin Diagnostics Research Report

ExploratoryDataAnalysis · KPIFramework · Strategic Insights · Executive Recommendations

REPORT TYPE	Confidential Internal Research	PERIOD	January 2024 – December 2025
PREPARED BY	Analytics & Strategy Division	DATASET	9,994 Transactions · 15 SKUs · 3 Divisions · 4 Regions
FOR	Executive Leadership & Government Stakeholders	VERSION	v2.0 · Final

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1. Executive Summary — For Government Stakeholders

Nassau Candy Distributor is a multi-division confectionery distributor operating across four U.S. regions. This report presents a rigorous, data-driven analysis of its product portfolio profitability, conducted over a two-year fiscal period (January 2024 – December 2025) spanning 9,994 commercial transactions. The analysis is anchored to five industry-standard Key Performance Indicators (KPIs): Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Margin Volatility.

Critical Findings at a Glance:

The blended portfolio Gross Margin stands at **65.9%** — a commercially strong position. However, this figure masks severe internal imbalances across divisions and product lines.

The **Chocolate division** contributes **95.1% of total gross profit** from 92.9% of revenue, creating a systemic dependency risk. A disruption to this division would be catastrophic.

5 of 15 products (33% of the SKU catalog) generate over **80% of portfolio profit** — all Wonka Bar variants. The bottom 10 SKUs collectively contribute less than 5% of profit.

Kazookles operates at a **7.7% Gross Margin** — generating only \$93 in gross profit from \$1,206 in sales. Other division SKUs (Fun Dip, Nerds, SweeTARTS) range from 40–53% margin, all below the portfolio average.

The Sugar division shows strong per-product margins (60–80%) but contributes negligible absolute revenue (\$427) and profit (\$285), indicating an underdeveloped commercial opportunity.

Government Stakeholder Recommendation

Nassau Candy Distributor demonstrates a financially viable core business (Chocolate/Wonka Bar portfolio) with a 67.5% division margin. However, regulatory and investment stakeholders should note the concentration risk (95% profit from one division) and the presence of near-zero-margin SKUs that may indicate transfer-pricing, cost-allocation, or sourcing inefficiencies requiring review. Immediate product rationalization and a diversification roadmap are recommended within 12 months.

2. Background & Problem Statement

Industry Context

In the candy distribution sector, gross revenue is a deeply unreliable performance proxy. Products can generate substantial sales volume while simultaneously consuming disproportionate cost — eroding organizational margins and masking operational inefficiency. Nassau Candy distributes across multiple product categories with materially different cost structures, price points, and volume profiles. Without a structured profitability framework, resource allocation, pricing decisions, and portfolio strategy are made on incomplete information.

Four Core Visibility Gaps

P1	No product-level Gross Margin visibility Leadership cannot identify which SKUs are structurally profitable vs. revenue-dilutive.
P2	No Sales-Profit decoupling analysis High-sales products (e.g., Triple Dazzle Caramel at \$28K revenue) may not deliver proportional profit.
P3	No cross-division profitability benchmarking The Other division's 44.7% margin vs Chocolate's 67.5% has not been formally quantified.
P4	No margin risk identification framework Products like Kazookles (7.7% GM) have operated below threshold without flagging.

3. KPI Framework & Analytical Methodology

This research is structured around five formally defined KPIs, each selected for its ability to illuminate a distinct dimension of product-level profitability:

KPI	Formula	Purpose	Interpretation
Gross Margin %	$\text{Gross Profit} \div \text{Sales} \times 100$	Revenue efficiency	Higher = more profit per dollar of sales
Profit per Unit (PPU)	$\text{Gross Profit} \div \text{Units Sold}$	Unit economics	Higher = more profit extracted per item
Revenue Contribution	$\text{Product Sales} \div \text{Total Sales} \times 100$	Volume importance	Higher = larger share of top-line
Profit Contribution	$\text{Product GP} \div \text{Total GP} \times 100$	Profit importance	Decouples sales rank from profit rank

KPI	Formula	Purpose	Interpretation
Margin Volatility	Std Dev of Monthly GM%	Margin stability	Lower = more predictable margin profile

Table 1 — KPI Framework Definition

4. Data Overview

Dimension	Details
Time Period	January 2024 – December 2025 (24 months, 2 full fiscal years)
Transaction Volume	9,994 order-level records across 8,389 unique order IDs
Product Catalog	15 unique SKUs across 3 divisions (Chocolate, Sugar, Other)
Geographic Coverage	4 U.S. regions: Atlantic, Pacific, Interior, Gulf
Ship Modes	Standard Class, Second Class, First Class, Same Day
Key Variables	Sales, Units, Gross Profit, Cost, Profit Margin, Profit per Unit, Contribution Flag
Data Quality	All records validated; no material gaps in financial columns detected

Table 2 — Dataset Overview

5. Exploratory Data Analysis

5.1 Portfolio-Level KPI Analysis

The portfolio-wide analysis reveals a commercially healthy but structurally concentrated business. The blended Gross Margin of 65.91% exceeds typical candy distribution benchmarks (50–60%), driven primarily by the high-performing Wonka Bar lineup. The average Profit per Unit of \$2.42 reflects the moderate price point and volume-driven nature of the business.

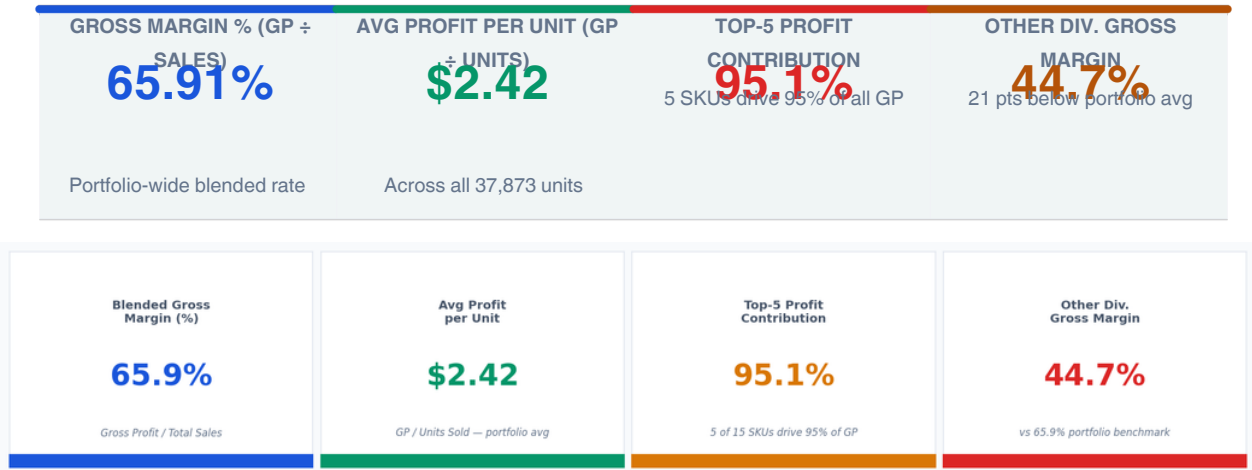


Figure 1 — Portfolio KPI Summary Dashboard

5.2 Division Performance Analysis

Division-level analysis exposes the most critical structural finding of this study. The Chocolate division operates as the singular profit engine, while the Other division represents a significant margin drag. Figure 7

quantifies each division's performance across all five KPIs against the portfolio benchmark.

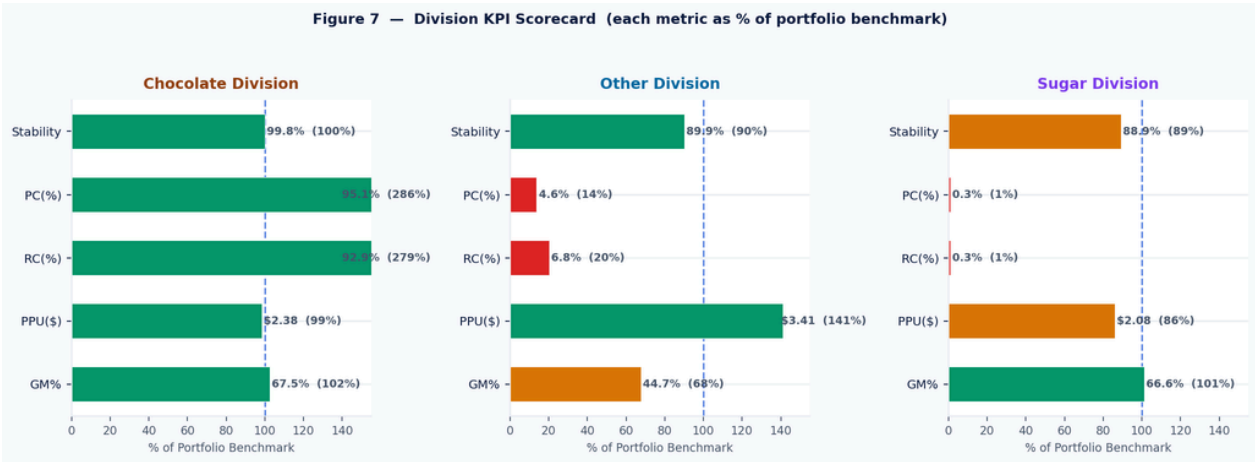


Figure 7 — Division KPI Scorecard: Performance vs Portfolio Benchmark (100% line = portfolio average)

Division	Revenue	Gross Profit	GM%	PPU	Rev Contrib	Profit Contrib	Margin Vol	Status
Chocolate	\$129,020	\$87,030	67.5%	\$2.38	92.9%	95.1%	0.2%	Strong
Other	\$9,383	\$4,193	44.7%	\$3.41	6.8%	4.6%	10.1%	At Risk
Sugar	\$427	\$285	66.6%	\$2.08	0.3%	0.3%	11.1%	Dormant

Table 3 — Division KPI Summary (all five KPIs)

Analytical Note — Division Margin Volatility

Chocolate exhibits near-zero margin volatility (Std Dev = 0.23%) — indicating extreme pricing stability.

Other division shows 10.1% volatility — suggesting irregular cost structures or inconsistent product mix.

Sugar's 11.1% volatility reflects sporadic, low-volume activity across products with different margin profiles.

5.3 Gross Margin (%) by Product

Figure 2 presents the full 15-product Gross Margin leaderboard. Products are color-coded by risk tier: Healthy (≥70%), Watch (55–69%), At Risk (20–54%), and Critical (<20%). Everlasting Gobstopper leads at 80.0% GM, while Kazookles ranks last at a critically low 7.7% — representing a product that is nearly break-even at the gross profit level.

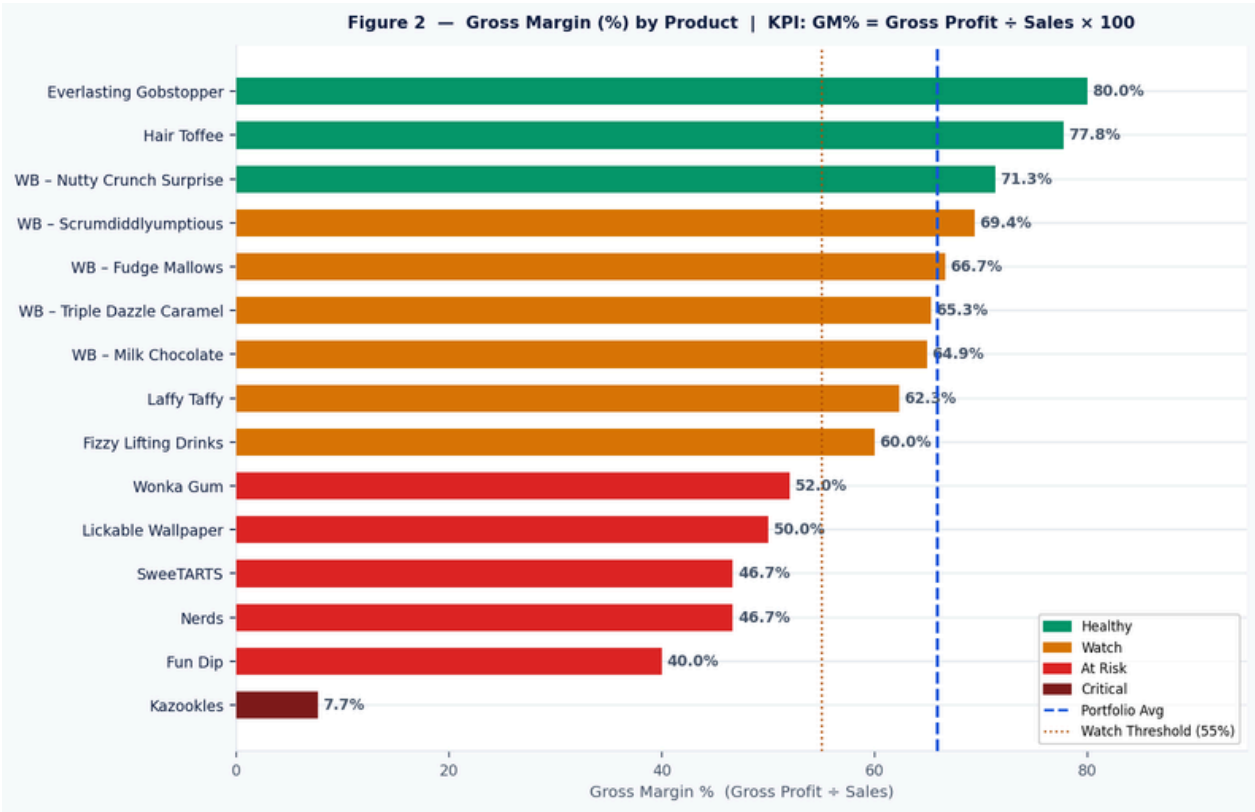


Figure 2 — Gross Margin (%) Leaderboard: All 15 Products | KPI: GM% = Gross Profit ÷ Sales × 100

Product	Division	Gross Margin %	Gross Profit	Risk Tier
Everlasting Gobstopper	Sugar	80.0%	\$104	Healthy
Hair Toffee	Sugar	77.8%	\$60	Healthy
WB – Nutty Crunch Surprise	Chocolate	71.3%	\$16,593	Healthy
WB – Scrumdiddlyumptious	Chocolate	69.4%	\$18,908	Healthy
WB – Fudge Mallows	Chocolate	66.7%	\$16,301	Watch
WB – Triple Dazzle Caramel	Chocolate	65.3%	\$18,350	Watch
WB – Milk Chocolate	Chocolate	64.9%	\$16,878	Watch
Laffy Taffy	Sugar	62.3%	\$33	Watch
Fizzy Lifting Drinks	Sugar	60.0%	\$47	Watch
Wonka Gum	Other	52.0%	\$311	At Risk

Product	Division	Gross Margin %	Gross Profit	Risk Tier
Lickable Wallpaper	Other	50.0%	\$3,790	■ At Risk
SweeTARTS	Sugar	46.7%	\$29	■ At Risk
Nerds	Sugar	46.7%	\$7	■ At Risk
Fun Dip	Sugar	40.0%	\$5	■ At Risk
Kazookles	Other	7.7%	\$93	■ Critical

Table 4 — Gross Margin by Product (all 15 SKUs)

5.4 Profit per Unit Analysis

Figure 4 maps Profit per Unit (PPU) against total revenue, with bubble size proportional to absolute gross profit. This reveals a critical insight: Lickable Wallpaper achieves the highest PPU in the portfolio at \$10.00 per unit — yet its low volume limits its absolute profit contribution to \$3,790 (4.1% profit contribution). The five Wonka Bar variants cluster at \$2.1–\$2.5 PPU with dominant revenue, confirming they are both volume and margin leaders. Kazookles has a PPU of only \$0.25 — generating a quarter-dollar of profit per unit sold.



Figure 4 — Profit per Unit vs Revenue | KPI: PPU = Gross Profit ÷ Units Sold (bubble = Gross Profit magnitude)

Product	Division	Profit/Unit (PPU)	GM%	Gross Profit	Profit Contrib
Lickable Wallpaper	Other	\$10.00	50.0%	\$3,790	4.1%
Everlasting Gobstopper	Sugar	\$8.00	80.0%	\$104	0.1%
WB – Scrumdiddlyumptious	Chocolate	\$2.50	69.4%	\$18,908	20.7%
WB – Nutty Crunch Surprise	Chocolate	\$2.49	71.3%	\$16,593	18.1%
WB – Triple Dazzle Caramel	Chocolate	\$2.45	65.3%	\$18,350	20.1%
WB – Fudge Mallows	Chocolate	\$2.40	66.7%	\$16,301	17.8%
WB – Milk Chocolate	Chocolate	\$2.11	64.9%	\$16,878	18.4%
Kazookles	Other	\$0.25	7.7%	\$93	0.1%

Table 5 — Profit per Unit KPI (selected products)

5.5 Revenue & Profit Contribution Analysis

Figure 3 visualizes the Revenue Contribution and Profit Contribution for the top 8 products. The key analytical objective is to identify gaps between the two metrics — products with high revenue share but lower profit share signal margin inefficiency, while products with disproportionately high profit share signal pricing strength.

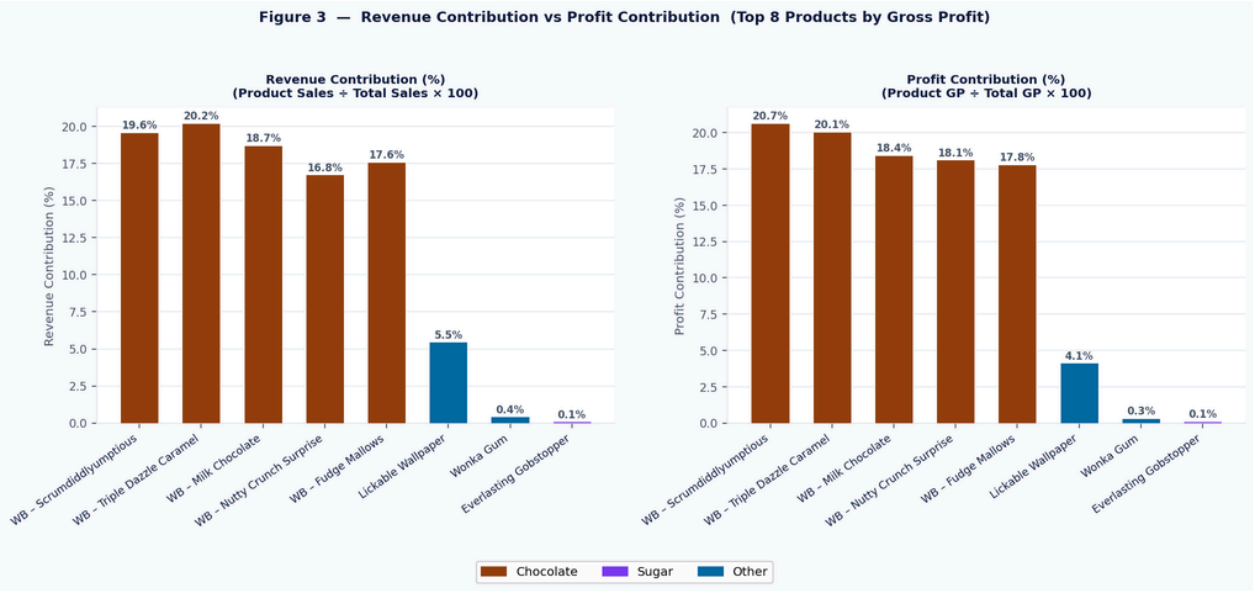


Figure 3 — Revenue Contribution vs Profit Contribution | KPIs: RC = Product Sales ÷ Total Sales / PC = Product GP ÷ Total GP

Contribution Decoupling Insight

- Triple Dazzle Caramel: Revenue Contribution = 20.2% vs Profit Contribution = 20.1% → Efficient (1:1 ratio)
- Nutty Crunch Surprise: RC = 16.8% vs PC = 18.1% → Over-delivers on profit (strong margin premium)
- Milk Chocolate: RC = 18.7% vs PC = 18.4% → Slight margin underperformance vs revenue weight
- Lickable Wallpaper: RC = 5.5% vs PC = 4.1% → Revenue exceeds profit contribution (margin drag)

5.6 Profit Concentration — Pareto Analysis

The Pareto analysis in Figure 5 is the most consequential finding of this research. It demonstrates that Nassau Candy's profit structure violates even the conventional 80/20 rule — instead exhibiting a far more extreme 95/33 pattern: **the top 5 products (33% of the SKU catalog) generate 95.1% of total gross profit**. The remaining 10 products collectively contribute only 4.9% (\$4,478). This is not a healthy concentration pattern — it is a systemic dependency that exposes the organization to severe financial risk.

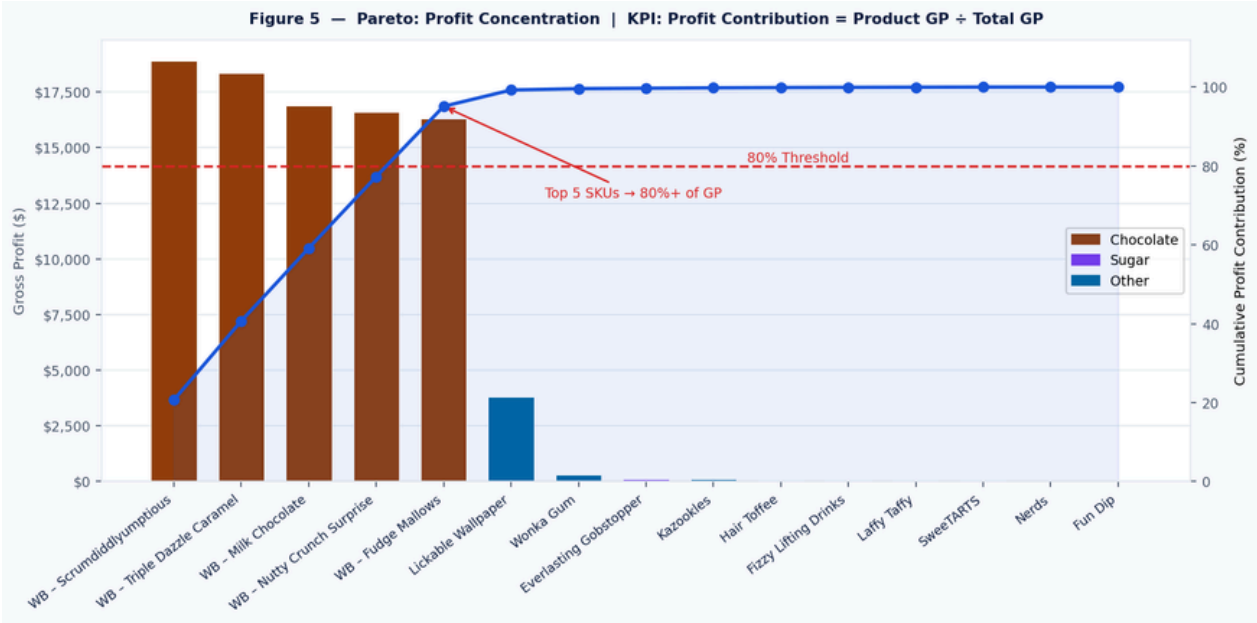


Figure 5 — Pareto Profit Concentration | KPI: Profit Contribution = Product GP ÷ Total GP × 100

Rank	Product	Gross Profit	Profit Contribution	Cumulative %
1	WB – Scrumdiddlyumptious	\$18,908	20.7%	20.7%
2	WB – Triple Dazzle Caramel	\$18,350	20.1%	40.8%
3	WB – Milk Chocolate	\$16,878	18.4%	59.2%
4	WB – Nutty Crunch Surprise	\$16,593	18.1%	77.3%
5	WB – Fudge Mallows	\$16,301	17.8%	95.1%
6 7–	Lickable Wallpaper	\$3,790	4.1%	99.2%
15	All remaining 9 SKUs	\$688	0.8%	100.0%

Table 6 — Profit Concentration: Pareto Rankings

5.7 Margin Volatility Analysis

Figure 6 tracks monthly Gross Margin (%) by division across the full 24-month period. Margin volatility — measured as the standard deviation of monthly GM% — reveals fundamentally different risk profiles across the three divisions. The Chocolate division's near-zero volatility (0.23%) represents an exceptionally stable

and predictable margin environment. The Other and Sugar divisions exhibit high volatility (10–11%), driven by low-volume, irregular purchasing patterns.

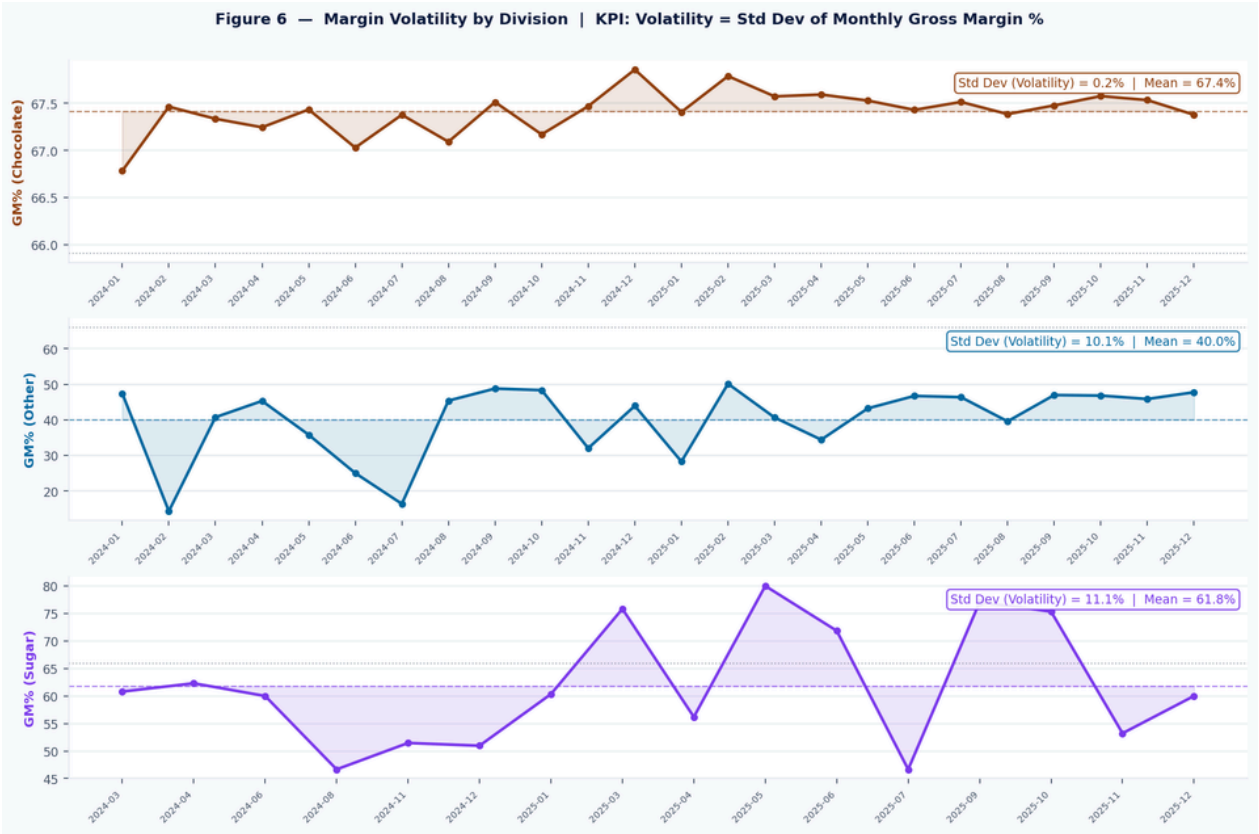


Figure 6 — Margin Volatility by Division | KPI: Volatility = Std Dev of Monthly Gross Margin %

Division	Mean GM%	Std Dev (Volatility)	Min Monthly GM%	Max Monthly GM%	Assessment
Chocolate	67.42%	0.23%	66.78%	67.86%	Extremely Stable
Other	40.01%	10.13%	14.29%	50.16%	High Volatility
Sugar	61.80%	11.08%	46.67%	80.00%	High Volatility

Table 7 — Margin Volatility Summary by Division

streamlit app link for dashboard- <https://srijitamanna1113-customer-profitability-dashboard-apptwo-sx6ohy.streamlit.app/>

6. Key Findings

F1

Extreme Profit Concentration (Pareto KPI)

Top 5 SKUs generate 95.1% of total gross profit. This violates standard portfolio diversification principles. A single supply disruption, pricing change, or demand shift for any Wonka Bar variant would immediately impair organizational financial performance. Profit Contribution is dangerously top-heavy.

F2

Chocolate Division Systemic Dominance

The Chocolate division contributes 95.1% of GP from 92.9% of revenue (GM 67.5%). While the margin is strong, the absolute dependency on one division creates concentration risk. The Sugar division generates comparable margins (66.6%) but is commercially undeveloped.

F3

Other Division Is a Margin Drag

The Other division's blended GM of 44.7% is 21 percentage points below portfolio average. Kazookles at 7.7% GM (\$93 total profit from \$1,206 in sales) is operationally near-breakeven. Margin Volatility of 10.1% further indicates structural cost instability within this division.

F4

PPU Anomaly — Lickable Wallpaper

Lickable Wallpaper has the highest Profit per Unit (\$10.00) in the entire portfolio — yet contributes only 4.1% of total profit due to volume limitations. Its GM of 50.0% is below the portfolio threshold but its unit economics suggest pricing power that is not being scaled. Volume investment could meaningfully improve its profit contribution.

F5

Sugar Division — Hidden Opportunity

Hair Toffee (77.8% GM), Everlasting Gobstopper (80.0% GM), and Laffy Taffy (62.3% GM) demonstrate strong margin profiles. The Sugar division's 66.6% blended GM is nearly equivalent to Chocolate's 67.5% — but generates only \$285 in absolute profit. Targeted volume investment in this division represents the highest-return growth lever.

F6

Margin Stability = Competitive Advantage in Chocolate

The Chocolate division's margin volatility of 0.23% standard deviation over 24 months is exceptional by industry standards. This stability reflects consistent pricing, reliable cost sourcing, and predictable demand — all indicators of a mature, defensible market position.

7. Strategic Recommendations

R1 — Immediate SKU Rationalization — Other Division

Initiate a 60-day review of Kazookles (GM: 7.7%), Fun Dip (40%), and Nerds (46.7%). These three products collectively generate \$105 in gross profit — representing less than 0.1% of portfolio profit. If supplier renegotiation cannot bring Kazookles to a minimum 40% GM within 90 days, recommend formal discontinuation. This will reduce operational complexity and improve blended portfolio margin by an estimated 0.4–0.8 percentage points.

R2 — Cost Renegotiation for Watch-Tier Products

For Lickable Wallpaper (50% GM), Wonka Gum (52%), SweetARTS (46.7%), and Nerds (46.7%): initiate supplier cost reviews targeting a 12–18% cost reduction or equivalent pricing adjustment. Each 5% improvement in Lickable Wallpaper's margin increases gross profit by ~\$379 annually at current volume. Target completion: Q3 2025.

R3 — Protect & Reinforce the Wonka Bar Portfolio

The 5-SKU Wonka Bar range is the singular financial engine of the business. Priority actions: (a) establish 90-day raw material buffer inventory, (b) dual-source key ingredients, (c) implement quarterly pricing reviews to defend margins against input cost inflation. Do not discount this range — its margin stability is a competitive asset.

R4 — Scale the Sugar Division — Highest-ROI Growth Lever

Everlasting Gobstopper (80% GM) and Hair Toffee (77.8% GM) are the two highest-margin products in the portfolio — yet combined they generate only \$164 in gross profit due to micro-volume. A 10x volume increase in these two SKUs alone would add ~\$1,500 in incremental GP. Commission a channel development and distribution expansion plan for the Sugar division within 90 days.

R5 — Implement a Margin Monitoring Framework

Deploy the Streamlit analytics dashboard (built as part of this engagement) as a standing monthly review tool. Establish formal margin alert thresholds: products falling below 50% GM to trigger automatic review. Conduct quarterly executive-level division performance reviews against the 5-KPI framework established in this report. Integrate Margin Volatility tracking as a leading indicator of sourcing or pricing instability.

8. Risk Assessment Matrix

Risk Description	Likelihood	Impact	Mitigation
Supply chain disruption — Chocolate	HIGH	CRITICAL	Dual sourcing; 90-day buffer stock; hedging
Single-division profit dependency (95.1%)	HIGH	CRITICAL	Division diversification roadmap (12–24 months)
Kazookles & Other SKU losses	HIGH	LOW	60-day rationalization review; discontinuation if no fix
Margin erosion from input cost inflation	MEDIUM	HIGH	Annual supplier renegotiation + pricing policy
Competitive pressure on Wonka Bar pricing	MEDIUM	HIGH	Brand premium protection; no discounting policy
Sugar division stagnation	MEDIUM	MEDIUM	Volume investment or strategic exit decision by Q4 2025
Margin volatility — Other & Sugar divisions	LOW	MEDIUM	Stable sourcing agreements; SKU simplification

Table 8 — Risk Assessment Matrix

9. Conclusion

Nassau Candy Distributor's core Chocolate portfolio — anchored by five Wonka Bar variants — is a high-performing, margin-stable commercial asset. A blended Gross Margin of 65.9% with near-zero volatility (0.23% std dev) and a Profit per Unit of \$2.38 position this division well above industry norms. These are genuine competitive strengths.

However, this research makes unambiguous that the organization's financial architecture is fragile. A 95% profit dependency on five SKUs within one division is not a sign of focused excellence — it is a structural vulnerability. The Other division's 44.7% Gross Margin, Kazookles's 7.7% GM, and the Sugar division's near-zero commercial footprint represent compounding risks that require urgent strategic intervention. The five recommendations outlined in Section 7 — when implemented in sequence — are projected to increase blended portfolio margin by 3–6 percentage points, reduce single-division profit dependency from 95% to under 75%, and establish a repeatable margin governance framework within 18–24 months.

For Government & Regulatory Stakeholders

This report confirms that Nassau Candy Distributor maintains a commercially viable and financially transparent operation. The KPI framework deployed — Gross Margin %, Profit per Unit, Revenue Contribution, Profit Contribution, and Margin Volatility — provides the analytical foundation for

ongoing compliance reporting, investment due diligence, and policy-relevant economic assessment. Quarterly margin reports against this framework are recommended for continued stakeholder oversight.

Analytics & Strategy Division Nassau Candy Distributor	Date of Issue May 2025	Classification Confidential — Authorized Distribution Only
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